

### **ACTIVITY: WATCH A YOUTUBE VIDEO ON KRUSKAL'S ALGORITHM**

Video Link: Kruskal's Algorithm (Video is by WIT Solapur)

<https://www.youtube.com/watch?v=Faf4ldcDrOw>

#### **Instructions:**

- Watch the YouTube video on Kruskal's Algorithm.
- Take notes on the key concepts, steps of the algorithm, and any examples provided in the video.

#### **Reflection Questions based on the video**

##### **Understanding the Basics:**

What is the primary purpose of Kruskal's Algorithm? How does it differ from other algorithms used for finding a Minimum Spanning Tree?

##### **Algorithm Steps:**

List the main steps involved in executing Kruskal's Algorithm. Why is it important to sort the edges before processing them?

##### **Cycle Detection:**

How does the union-find data structure assist in detecting cycles during the execution of Kruskal's Algorithm? Explain the Find and Union operations.

##### **Practical Applications:**

In what real-world scenarios can Kruskal's Algorithm be effectively applied?

Provide at least two examples.

##### **Algorithm Efficiency:**

Discuss the time complexity of Kruskal's Algorithm. Under what circumstances might this algorithm be more efficient compared to other MST algorithms like Prim's Algorithm?

##### **Personal Insight:**

Reflect on your understanding of Kruskal's Algorithm after watching the video.

What concepts or steps did you find most challenging, and how would you approach them differently in the future?

##### **Submission**

Write your answers to the reflection questions and any additional notes from the video.